WELCOME

PROCESS CAPABILITY

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Common Questions for Investigating an Out-of-Control Process

- Are there differences in the measurement accuracy of instruments / methods used?
- Are there differences in the methods used by different personnel?
- Is the process affected by the environment, e.g. temperature/humidity?
- Has there been a significant change in the environment?

Common Questions for Investigating an Out-of-Control Process

- Is the process affected by predictable conditions such as tool wear?
- Were any untrained personnel involved in the process at the time?
- Has there been a change in the source for input to the process such as a new supplier or information?
- Is the process affected by employee fatigue?

Common Questions for Investigating an Out-of-Control Process

- Has there been a change in policies or procedures such as maintenance procedures?
- Is the process frequently adjusted?
- Did the samples come from different parts of the process? Shifts? Individuals?
- Are employees afraid to report “bad news”?

Estimating Sigma

The Control Chart Method for Variables Data

1. Construct the control chart and remove all special causes.

   <u>NOTE</u>: special causes are “special” only in that they come and go, not because their impact is either “good” or “bad”.

Estimating Sigma

The Control Chart Method for Variables Data

2. Estimate the standard deviation. The approach used depends on whether a R or S chart is used to monitor process variability.

$$\hat{\sigma} = \bar{R} / d_2 \qquad \hat{\sigma} = \bar{S} / c_4$$

Process Capability

What it is

- **Capability** is a process which gives a set of indexes which tell us how capable we are in meeting customer specifications, in the short term and in the long term.

Process Capability

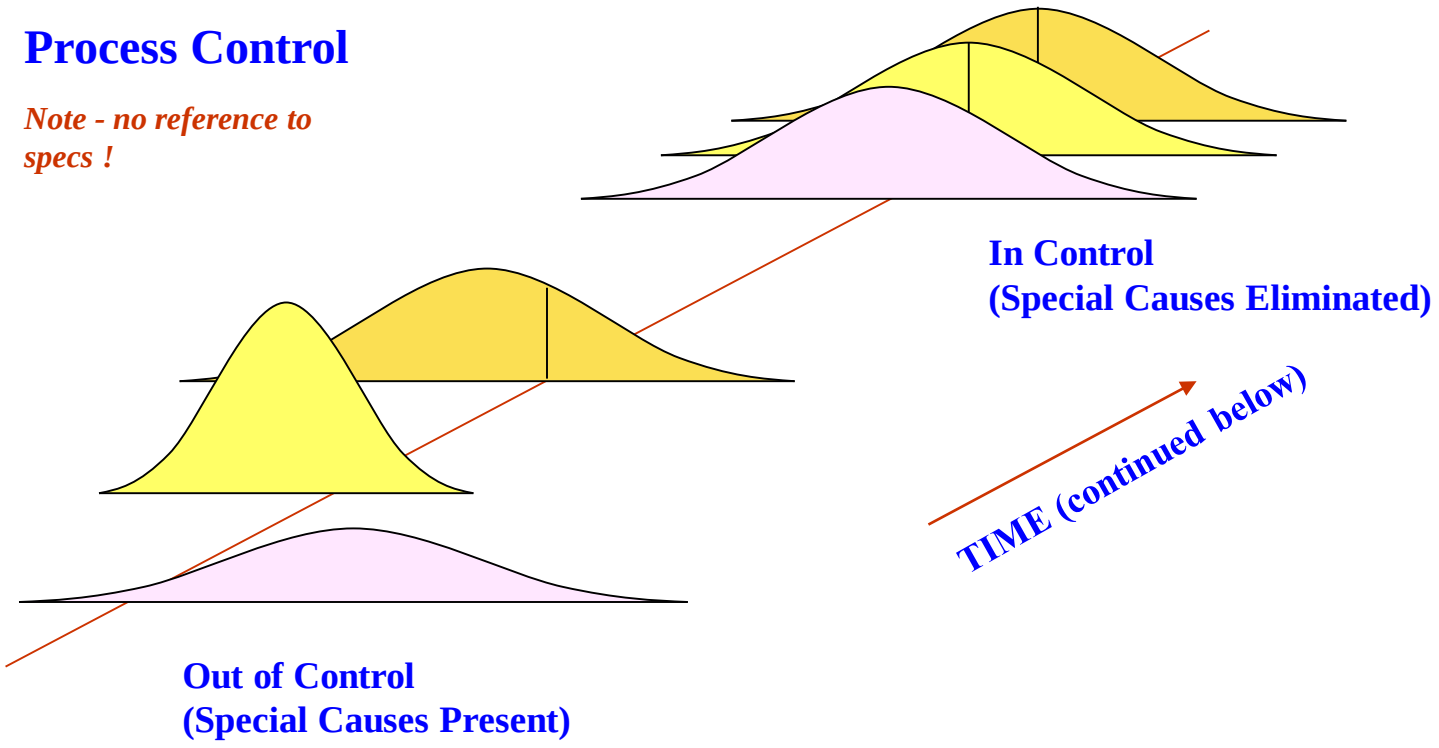
What it is

A simple view of Process Control and Process Capability

Process control - refers only to the "voice of the process" – looking at the process using an agreed performance measure to see whether the process forms a stable distribution over time.

Process Capability

Process Control



Process Capability

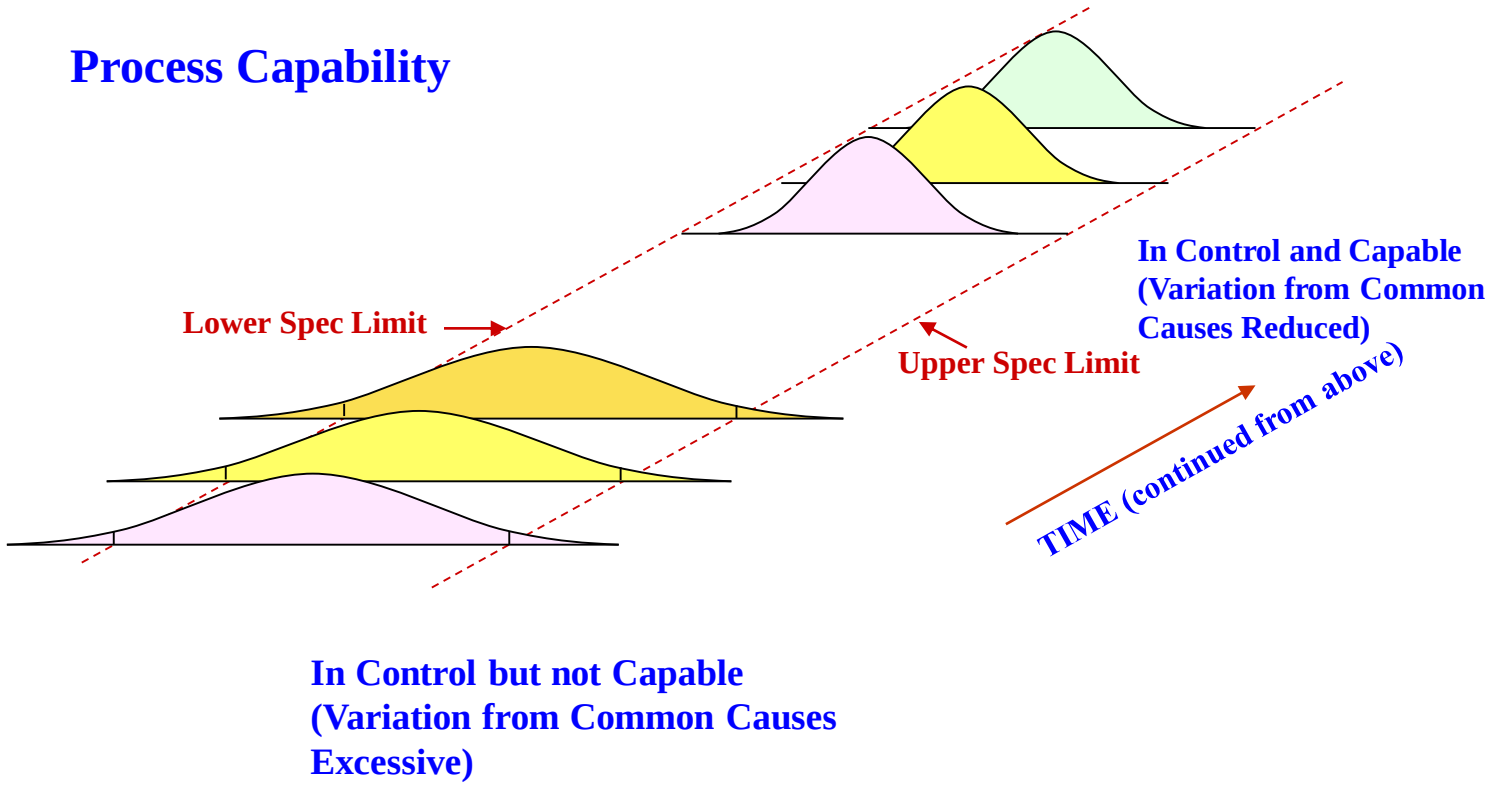
What it is

A simple view of Process Control and Process Capability

Process capability - measures the “goodness of a process” - comparing the voice of the process with the “voice of the customer”. The voice of the customer here is the specification range (tolerance) and/or the nearest customer specification limit.

Process Capability

Process Capability



Process Capability

Why Processes Fail?

- Process variation (spread) is too large
- Process average is not properly centered
- Process average is not properly centered and Process Variation is too large

Process Capability

Objective ?

- **The objective** is to get as close to the theoretical best that your process can achieve by eliminating special causes of variation,

- so that only common (natural) causes are acting on the process, and then to reduce these to a minimum, whenever possible.

Process Capability

Measurement

– We are often required to compare the output of a stable process with the process specifications and make a statement about how well the process meets specification.

–To do this we compare the natural variability of a stable process with the process specification limits.

Process Capability

- Assume that a process involves a quality characteristic that follows a normal distribution with mean μ, and standard deviation, σ. The upper and lower natural tolerance limits of the process are

- UNTL = $\mu + 3\sigma$
- LNTL = $\mu - 3\sigma$

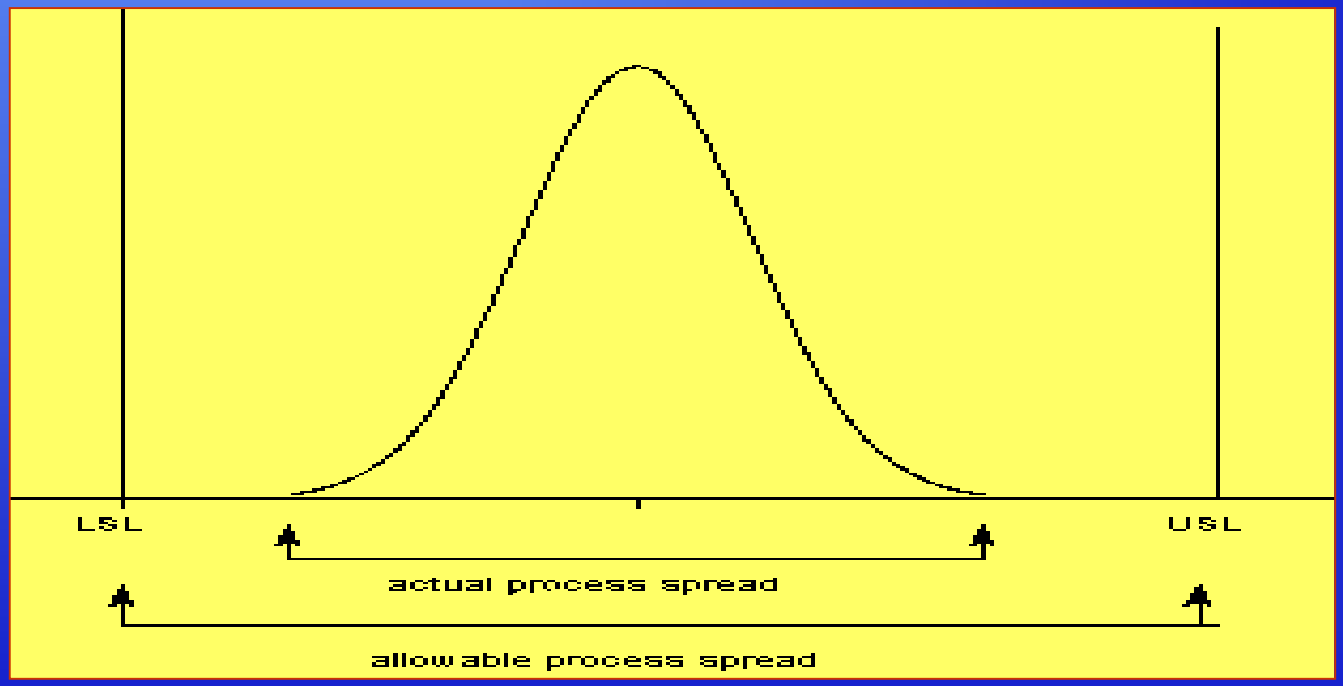
Actual Process Spread = UNTL – LNTL

Allowable Process Spread = USL - LSL

Process Capability

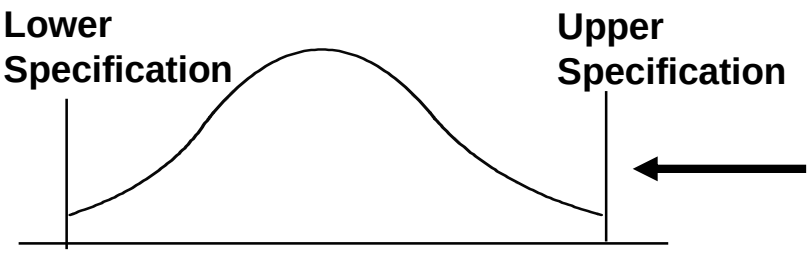
Measurement

– A capable process is one where almost all the measurements fall inside the specification limits. This can be represented pictorially by the plot below:



Actual Process Spread < Allowable Process Spread

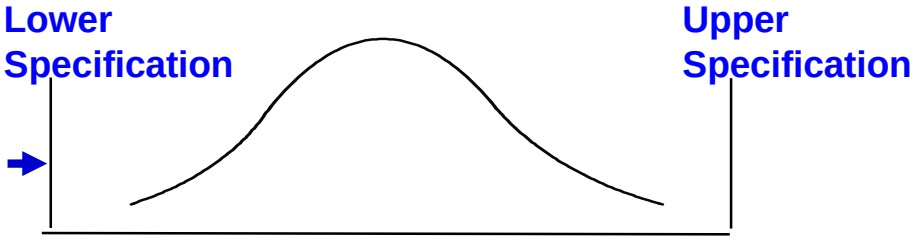
Process Capability



Actual Process Spread = Allowable Process Spread

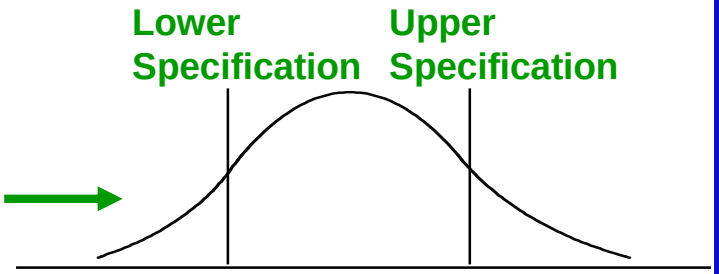
A. Process variability matches specifications

Actual Process Spread < Allowable Process Spread



B. Process variability well within specifications

Actual Process Spread > Allowable Process Spread

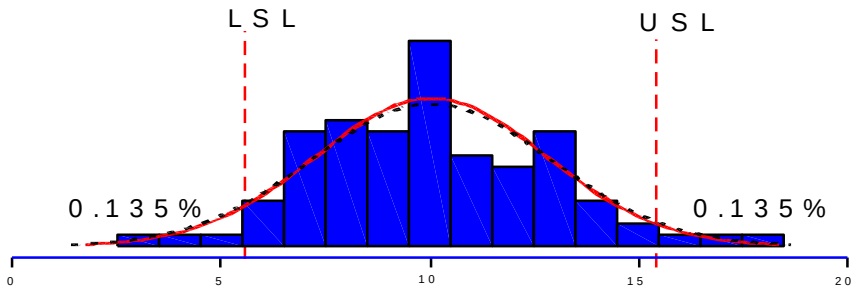


C. Process variability exceeds specifications

Process Capability

Process Capability for Bilateral Specification

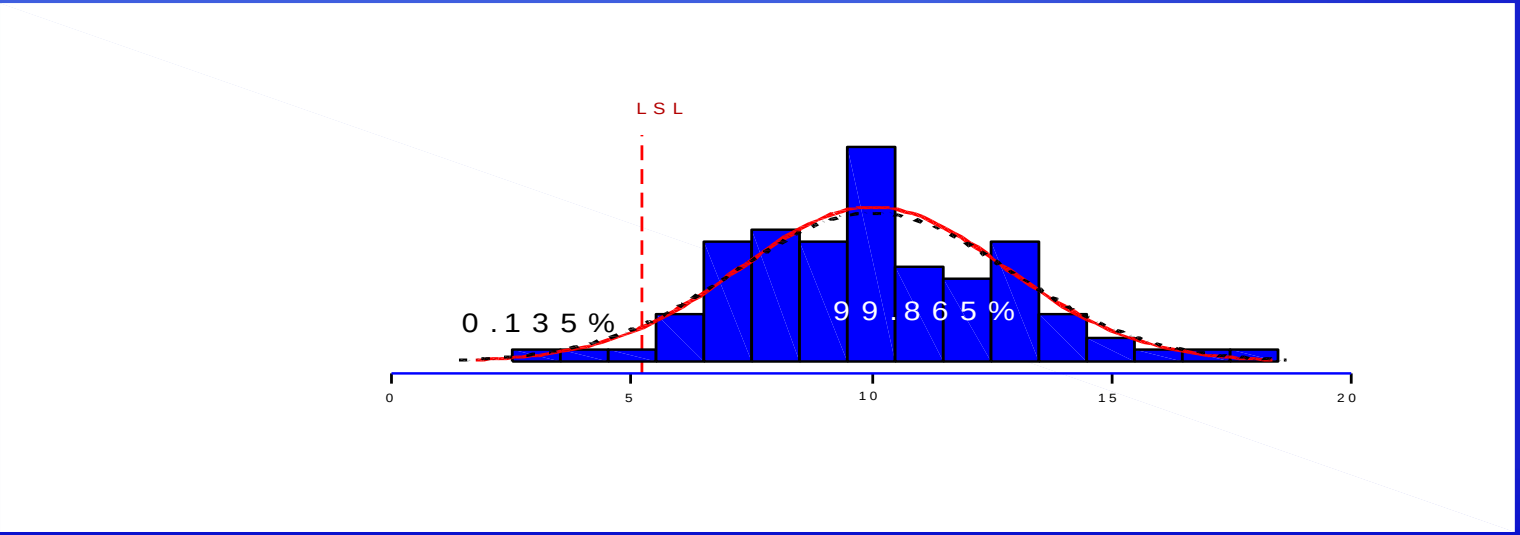
A process producing a characteristic with a bilateral specification meets the minimum requirement of capability when it is stable, and has no more than 0.135 percent of its output for this characteristic outside either specification limit.



Process Capability

Process Capability for Unilateral Specification

A process producing a characteristic with a unilateral specification meets the minimum requirement of capability when it is stable, and has no more than 0.135 percent of its output for this characteristic outside the single specification limit.



Process Capability Indices

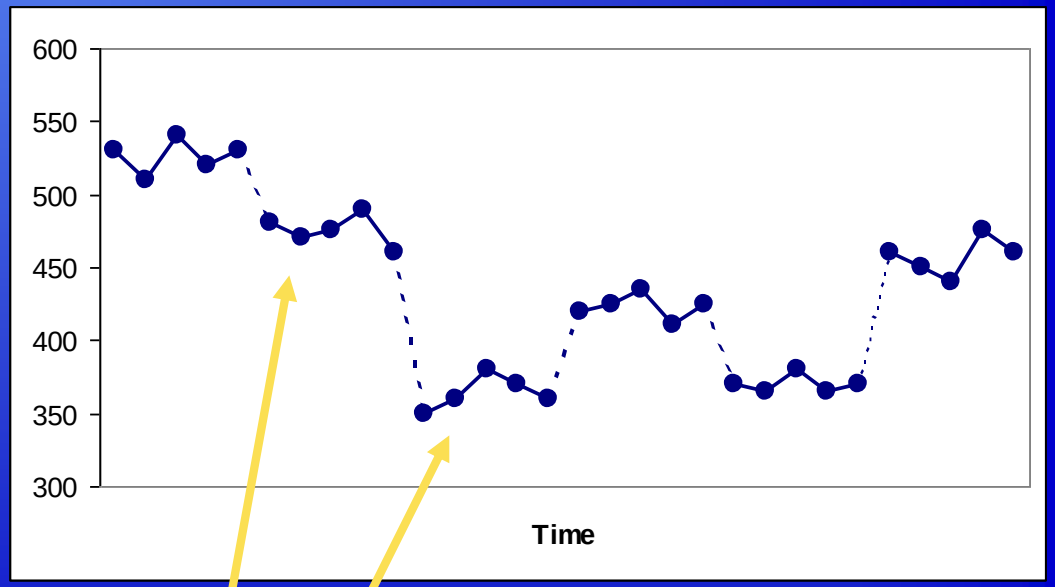
			1 CAPABILITY Short Term In Control Pooled std dev	2 PERFORMANCE Long Term Not in Control Total / overall std dev
a	How much variability	compares spec range (tolerance) to process width	C_p	P_p
b	How centred	how close process centre is to nearest spec limit	C_{pk}	P_{pk}

C_p or ENTITLEMENT is the theoretical best that a process can achieve.

Explaining the Terms

			1 CAPABILITY Short Term In Control Pooled std dev	2 PERFORMANCE Long Term Not in Control Total / overall std dev
a	How much variability	compares spec range (tolerance) to process width	C_p	P_p
b	How centred	how close process centre is to nearest spec limit	C_{pk}	P_{pk}

long term



short term

Explaining the Terms

			1 CAPABILITY Short Term In Control Pooled std dev	2 PERFORMANCE Long Term Not in Control Total / overall std dev
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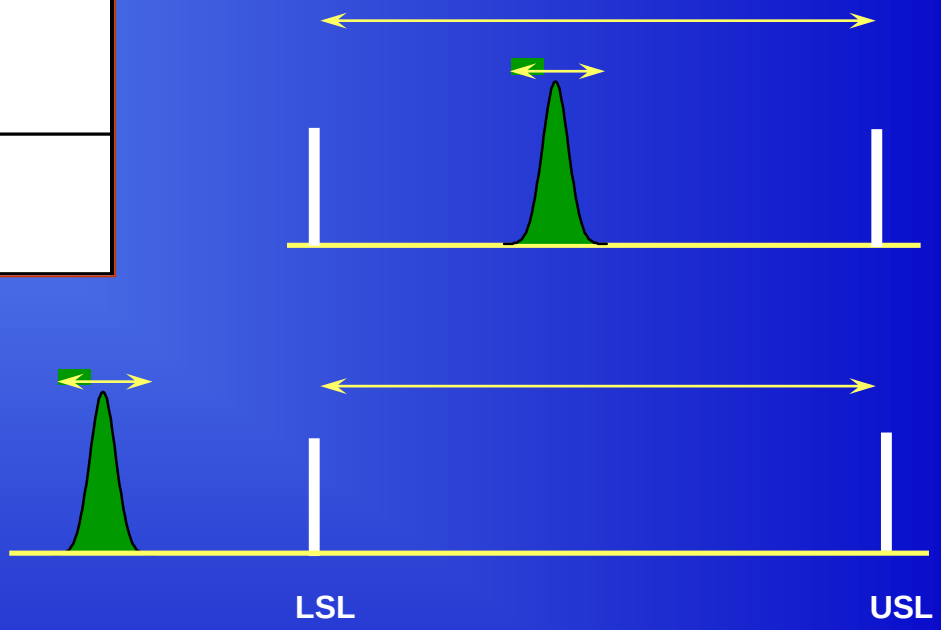
b	How centred	how close process centre is to nearest spec limit	C_{pk}	P_{pk}



C_p, P_p compares specification range (tolerance) to process width regardless of where the process is centred.

Process Capability

C_p, P_p compares specification range (tolerance) to process width regardless of where the process is centred.

It is the ratio of the specification width to
6 x process standard deviation.

$$C_p \text{ (Process Capability Index)} = \frac{USL-LSL}{6s_P}$$

$$P_p \text{ (Process Performance Index)} = \frac{USL-LSL}{6s_T}$$

s_P = pooled standard deviation s_T = total standard deviation

Explaining the Terms

		1 CAPABILITY Short Term In Control Pooled std dev	2 PERFORMANCE Long Term Not in Control Total / overall std dev
a How much variability	compares spec range (tolerance) to process width	C_p	P_p
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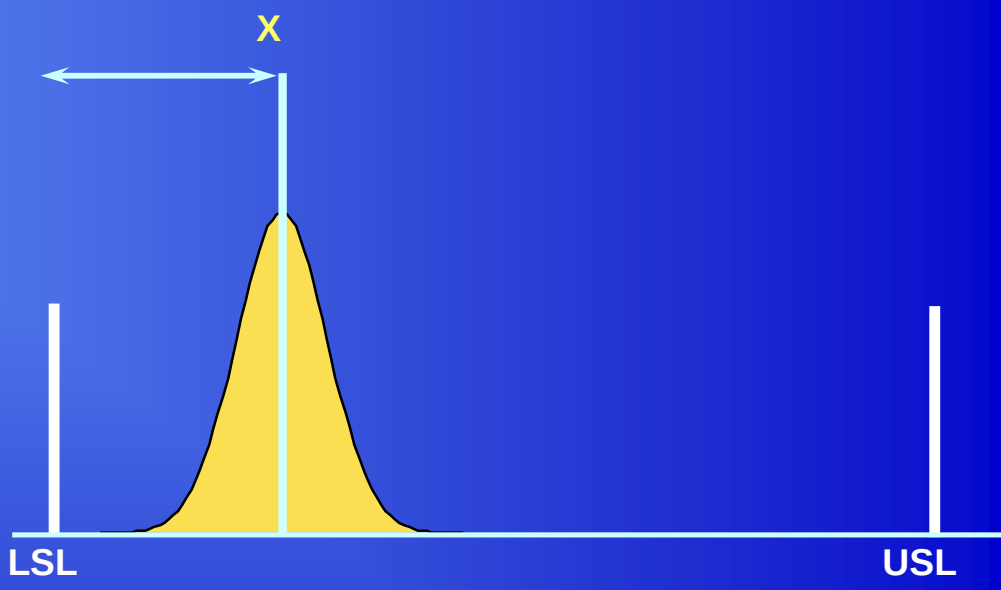


Cpk, Ppk measure how close the process centre is to the nearest customer spec.

It is (difference between the process average and the nearest specification) (3 x the process standard deviation).

Process Capability

$$C_{pkl} = \frac{X\text{-}LSL}{3s_P}$$

$$C_{pku} = \frac{USL\text{-}X}{3s_P}$$

$$C_{pk} = \min\{C_{pkl}, C_{pku}\}$$

$$P_{pkl} = \frac{X\text{-}LSL}{3s_T}$$

$$P_{pku} = \frac{USL\text{-}X}{3s_T}$$

$$P_{pk} = \min\{P_{pkl}, P_{pku}\}$$

Example

Humidity level in the foundry shop is expected to be between 60% and 80%. Random samples taken yield the following results:

60, 75, 70, 85, 60, 65,76,80, 70, 75

Average of the above ten values: 71.6,

SD = 8.24

Example

$$C_p = \frac{USL - LSL}{6\sigma} = \frac{80 - 60}{6(8.24)} = 0.40$$

$$C_{pkL} = \frac{\bar{x} - LSL}{3\sigma} = \frac{71.6 - 60}{3(8.24)} = 0.47$$

$$C_{pkU} = \frac{USL - \bar{x}}{3\sigma} = \frac{80 - 71.6}{3(8.24)} = 0.34$$

It is required to have at least 1.0, so this is not a good process. If possible, reduce the variability or/and center the process.

$$C_{pk} = \min\,(C_{pkU}, C_{pkL}) = \min\,(0.34, 0.47) = 0.34$$

Process Capability

The C_{pk} can also be expressed as $C_{pk} = C_p(1-k)$, where k is a scaled distance between the midpoint of the specification range, m, and the process mean, μ

$$k = \frac{|m - \mu|}{(USL - LSL)/2}, \qquad 0 \le k \le 1$$

$$\hat{C}_{pk} = \hat{C}_p(1 - \hat{k}) \qquad \hat{C}_{pk} \le \hat{C}_p$$

Example

For a certain process the USL = 18 and the LSL = 6.
The observed process average, $\bar{x} = 14$, and the standard deviation, s = 2.

$$C_p = \frac{USL - LSL}{6\sigma} = \frac{18 - 6}{6(2)} = 1.0$$

This means that the process is capable as long as it is located at the midpoint, m = (USL + LSL)/2 = 12.

Example

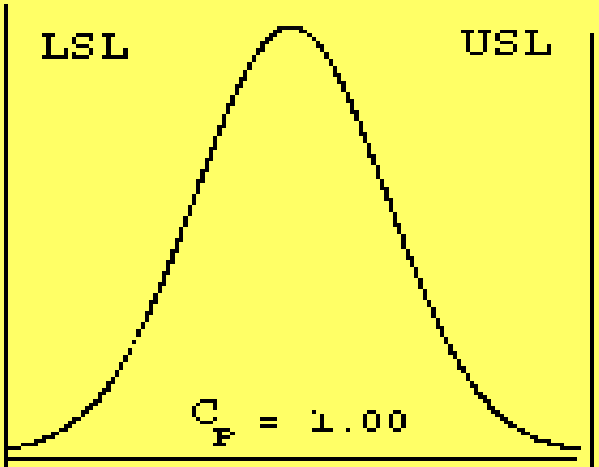
But it doesn't, since $\bar{x}$ = 14. The k factor is found by

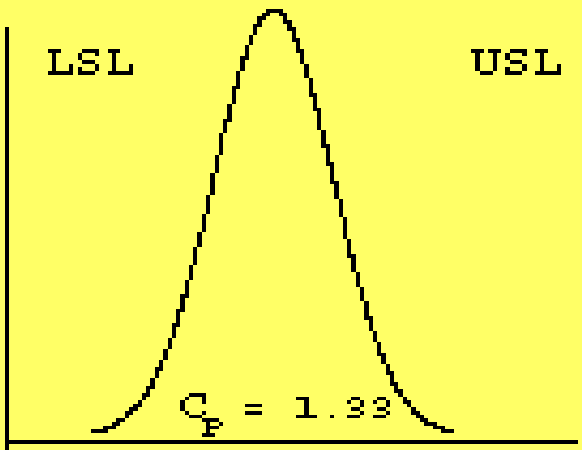
$$k = \frac{\left| m - \bar{x} \right|}{(USL - LSL)/2} = \frac{2}{6} = 0.333$$

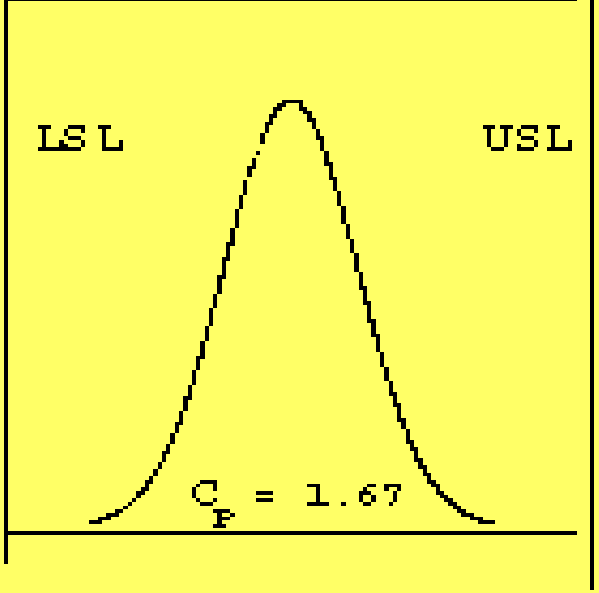
$$C_{PK} = C_P(1-k) = 0.667$$

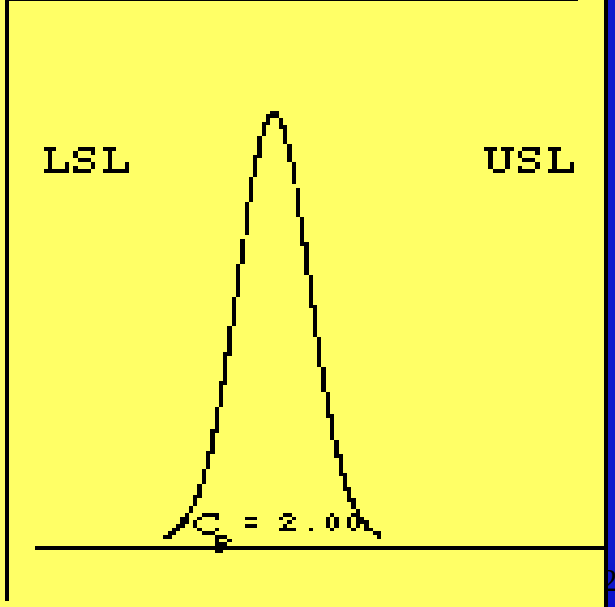
It is required to have C_{pk} at least 1.0, so this is not a good process. If possible, reduce the variability or/and center the process.

Process Capability Index









Translating capability into "rejects"

USL - LSL	6σ	8σ	10σ	12σ
C_p	1.00	1.33	1.66	2.00
Rejects	2700 ppm	64 ppm	0.6 ppm	2 ppb
% of spec used	100	75	60	50

Process Capability

use ***as a rule of thumb*** the following chart

	Cp	Cpk	Pp	Ppk
Red (Bad)	< 1.00	< 1.00	< 1.33	< 1.33
Yellow (OK)	1.00 - 1.33	1.00 - 1.33	1.33 - 1.67	1.33 - 1.67
Green (Good)	> 1.33	> 1.33	> 1.67	> 1.67

Process Capability

How it helps

A Process Capability study is the end result of the analysis done on process output data.

A Process Capability study reflects what our customer sees.

It indicates what our short term and long term situation looks like.

Process Capability

How it helps

Recognise however that our processes need to be stable (in statistical control) as well as capable.

Before anything, we must have faith in our measurements, hence the need for a measurement system analysis.

Process Capability

Major uses of data from a process capability analysis

1. Predicting how well the process will hold the tolerances.
2. Assisting product developers/designers in selecting or modifying a process.
3. Assisting in Establishing an interval between sampling for process monitoring.

Process Capability

Major uses of data from a process capability analysis

4. Specifying performance requirements for new equipment.
5. Selecting between competing vendors.
6. Planning the sequence of production processes when there is an interactive effect of processes on tolerances
7. Reducing the variability in a manufacturing process.

Process Capability

–**Setting Tolerances on Assemblies and Components**

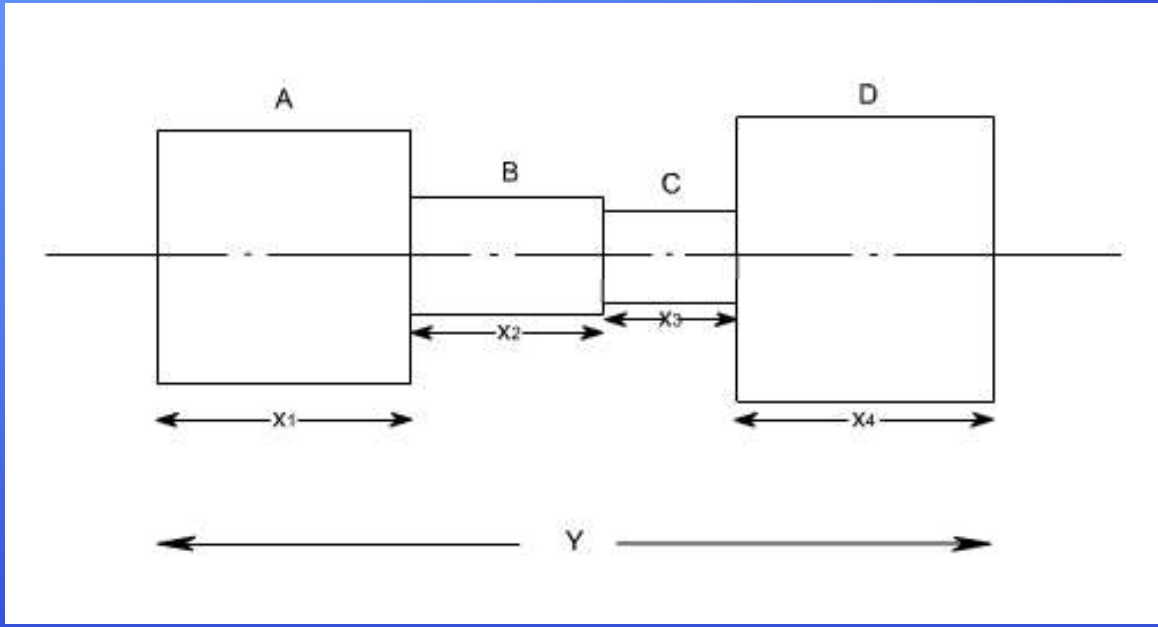
Assumption

–The process making assemblies and components are barely capable

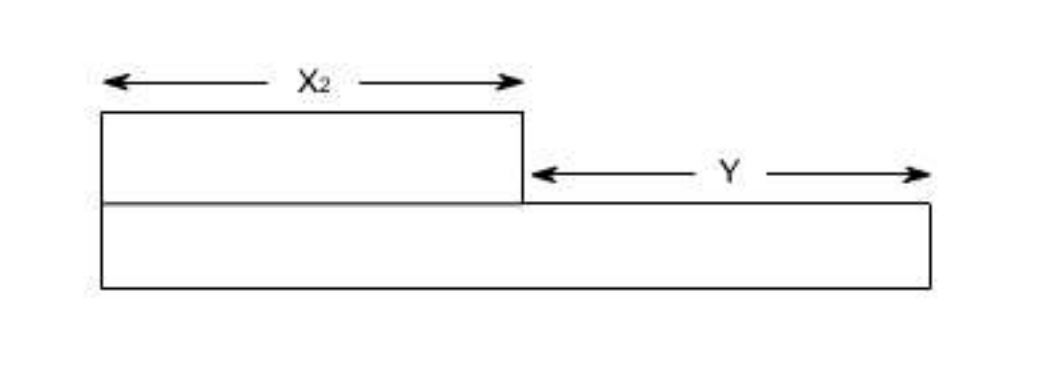
–i.e. $6\sigma = USL - LSL$

–or $Cp = 1$

Process Capability



$$Y = X1 + X2 + X3 + X4$$



$$Y = X1 - X2$$

Process Capability

In general (LC)

$Y = a_1X_1+a_2X_2+\ldots..+a_kX_k$

$$= \sum_{i=1}^{k} a_i x_i$$

–X_i (component dimension) & Y (assembly dimension) are random variables

–$X_i \rightarrow$ with mean μ_i and SD σ_i

If component dimension are independent

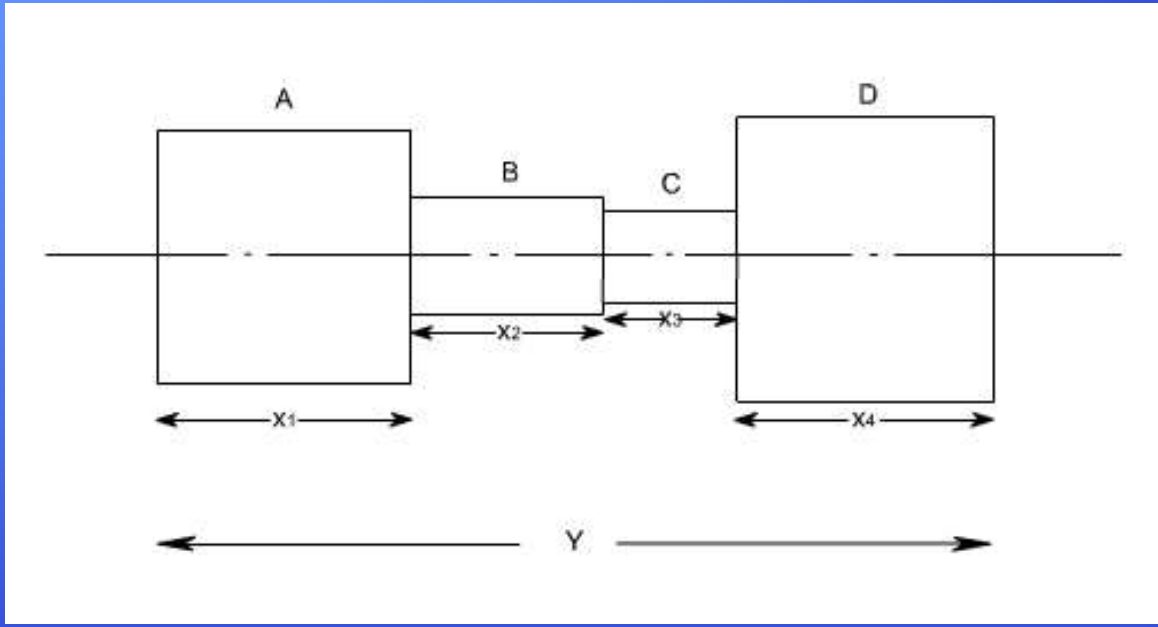
$$\mu_y = \sum_{i=1}^{k} a_i \mu_i$$

$$\text{Vari}(y) = \sigma_y^2 = \sum_{i=1}^{k} a_i^2 \, \text{var}(x_i) = \sum_{i=1}^{k} a_i^2 \sigma_i^2$$

μ_i = mean of X_i (variable)

a_i = constant.

Process Capability

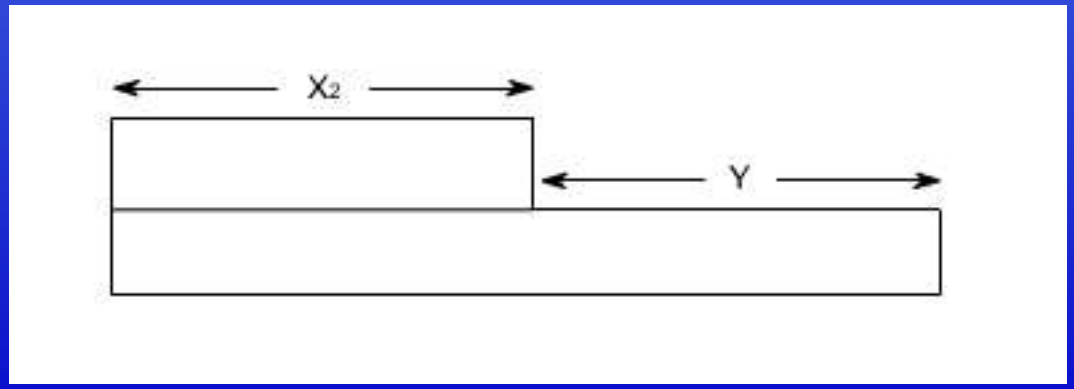


$$Y = X1 + X2 + X3 + X4$$

$$\mu_y = \mu_1 + \mu_2 + \mu_3 + \mu_4$$

$$\sigma_y = \sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \sigma_4^2}$$

$$NT \quad onY \qquad \mu_y \pm 3\sigma_y$$



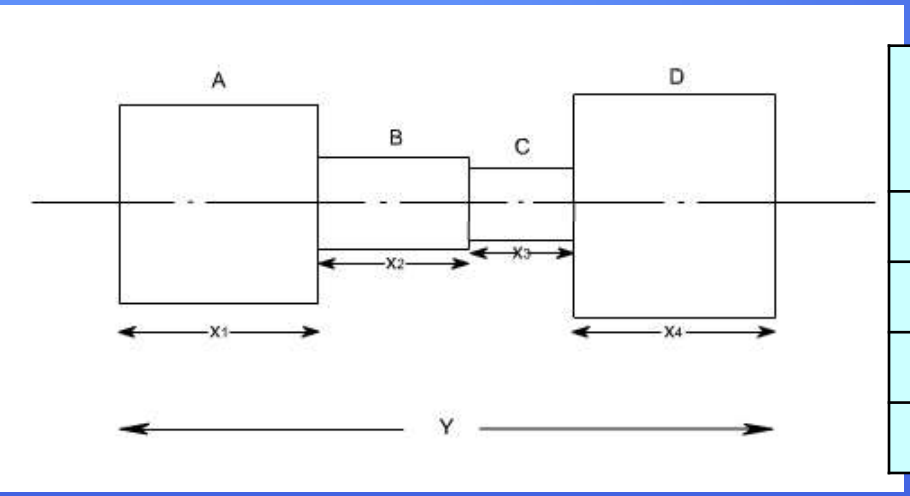
$$Y = X1 - X2$$

$$\mu_y = \mu_1 - \mu_2$$

$$\sigma_y = \sqrt{\sigma_1^2 + \sigma_2^2}$$

$$NT \quad onY \qquad \mu_y \pm 3\sigma_y$$

Process Capability



Components	Mean Length	Tolerances
A	2	2±0.3
B	5	5±0.2
C	6	6±0.2
D	7	7±0.1

–Design specifications of assembly length are 20±0.3

–Find the NTL for the assembly length.

–Comp. A $\sigma_1 = (2.3-1.7)/\ 6 = 0.1$ $\mu_1 = 2$

– B $\sigma_2 = (5.2-4.8)/\ 6 = 0.067$ $\mu_2 = 5$

– C $\sigma_3 = 0.067$ $\mu_3 = 6$

– D $\sigma_4 = 0.033$ $\mu_4 = 7$

Process Capability

–Comp. A σ_1 = (2.3-1.7)/ 6 = 0.1 $\mu_1 = 2$

– B σ_2 = (5.2-4.8)/ 6 = 0.067 $\mu_2 = 5$

– C σ_3 = 0.067 $\mu_3 = 6$

– D σ_4 = 0.033 $\mu_4 = 7$

$$\mu_y = \mu_1 + \mu_2 + \mu_3 + \mu_4$$

$$= 2 + 5 + 6 + 7 = 20$$

$$\sigma_y = \sqrt{\sigma_1^{\ 2} + \sigma_2^{\ 2} + \sigma_3^{\ 2} + \sigma_4^{\ 2}}$$

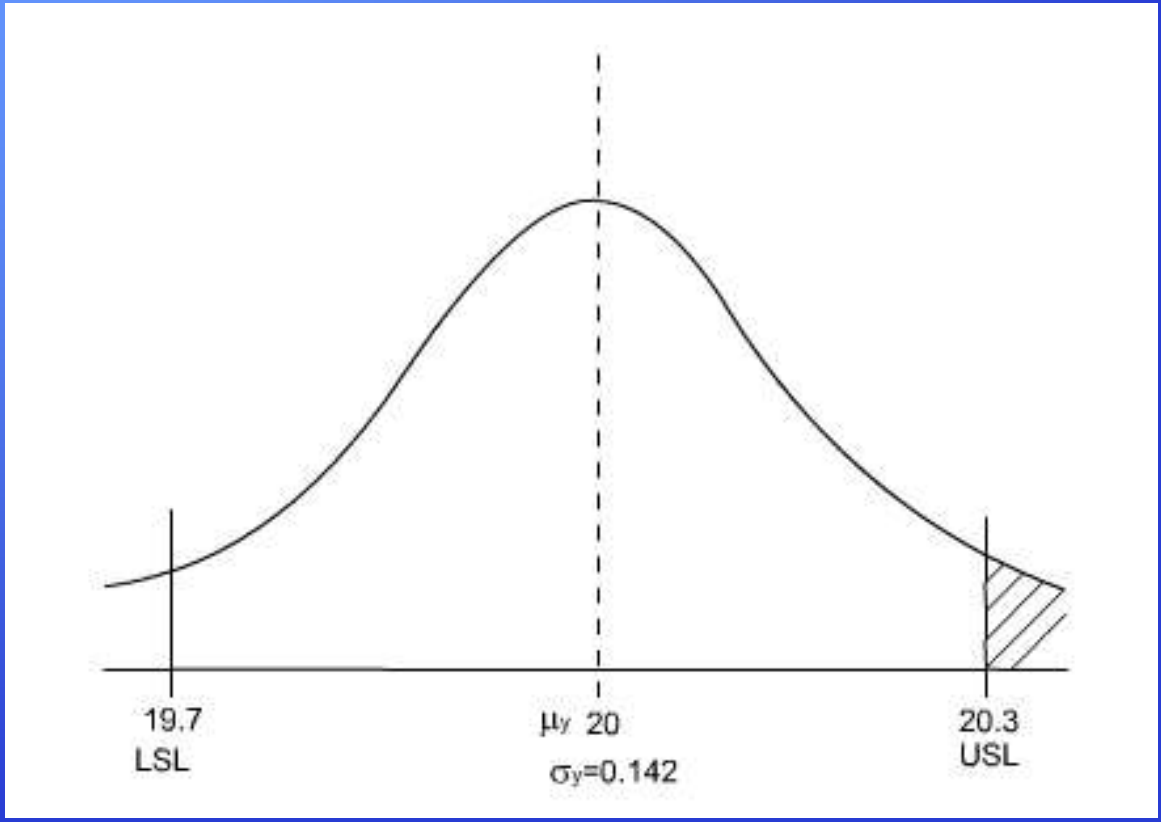
$$\sigma_y = \sqrt{0.1^2 + 0.67^2 + 0.67^2 + 0.33^2}$$

$$\sigma_y = 0.142$$

$$NT \quad onY \qquad \mu_y \pm 3\sigma_y$$

$$20 \pm 3(0.142) = (19.574, 20.426)$$

Process Capability



Z_{USL} = (20.3 – 20)/ 0.142 = 2.11
Z_{LSL} = (19.7 – 20)/ 0.142 = -2.11

$$\left.\begin{cases} \Delta USL = .0174 \\ \Delta LSL = .0174 \end{cases}\right\} \text{Total Area} = .0348 = 3.48\%$$

Process Capability

Components	Mean Length
A	2
B	5
C	6
D	7

Tolerance on Individual Components.

–Design specifications of assembly length are 20±0.3

–Find the NTL for all the components

$$\mu_y = \mu_1 + \mu_2 + \mu_3 + \mu_4$$

$$\sigma_y = \frac{20.3 - 19.7}{6} = 0.1$$

$$\sigma_y^{\ 2} = \sigma_1^{\ 2} + \sigma_2^{\ 2} + \sigma_3^{\ 2} + \sigma_4^{\ 2} = 0.01$$

assuming equal distributi on of variance on all components

$$\sigma_y^{\ 2} = 4\sigma^2$$

$$\sigma^2 = \frac{\sigma_y^{\ 2}}{4} = \frac{.01}{4} = 0.0025$$

$$A \rightarrow 2 \pm 3x0.05$$

$$B \rightarrow 5 \pm 3x0.05$$

$$C \rightarrow 6 \pm 3x0.05$$

$$D \rightarrow 7 \pm 3x0.05$$

A = (1.85, 2.15)

B = (4.85, 5.15)

C = (5.85, 6.15)

D = (6.85, 7.15)

THANKS

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